

Things To Do



**A PROJECT WORK SUBMITTED TO THE FACULTY OF
INSTITUTE OF SCIENCE AND TECHNOLOGY**

**ACADEMIA INTERNATIONAL COLLEGE
GWARKO, LALITPUR,**

**FOR THE AWARD OF
BACHELOR OF SCIENCE (B.Sc.) IN COMPUTER SCIENCE AND
INFORMATION TECHNOLOGY (CSIT)**

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APRIL 2025

RECOMMENDATION

This is to recommend that **Rakesh Joshi and Pratyusha Shrestha**, (79010887, 5-2-1078-15-2022 & 79010885, 5-2-1078-13-2022), have carried out project work entitled "**Things To Do**" for the requirement to the project work in Bachelor of Science (B.Sc.) degree in Computer Science and Information Technology (CSIT) under my/our supervision in the Department of Institute of Science and Technology (IoST), Academia International College, Institute of Science and Technology (IoST), Tribhuvan University (T.U.), Nepal.

To my/our knowledge, this work has not been submitted for any other degree.

He/She has fulfilled all the requirements laid down by the Institute of Science and Technology (IoST), Tribhuvan University (T.U.), Nepal for the submission of the project work for the partial fulfillment of the Bachelor of Science in Computer Science and Information Technology (B.Sc. CSIT) degree.

Supervisor:

Amit Chapagain

14th, OCT 2025

ACKNOWLEDGEMENTS

First and foremost, we would like to express my deepest gratitude to our supervisor, **Amit Chapagain**, for his invaluable guidance, constant encouragement, and insightful suggestions throughout the development of this project. His expertise and constructive feedback were instrumental in shaping these Things To Do into a functional and user-friendly application.

We are equally grateful to the faculty members of the Department of **IoST, Academia International College**, for providing me with the necessary resources and a conducive learning environment. The knowledge and skills We acquired during our coursework greatly contributed to the successful completion of this project.

We would also like to acknowledge our friends and classmates for their moral support, productive discussions, and helpful feedback during the testing phase of Things To DO. Their suggestions helped improve the app's functionality and user experience.

Lastly, we extend my heartfelt appreciation to our family for their unwavering support, patience, and encouragement throughout our academic journey. Their belief in our abilities motivated us to persevere and complete this project successfully.

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ABSTRACT

“Things To Do” is a task management application designed to help users organize daily activities efficiently. This project implements core functionalities including task creation, categorization, priority setting, and completion tracking. Developed using modern web technologies (HTML, CSS, JavaScript), the application features an intuitive user interface with responsive design for cross-platform accessibility. The system architecture follows MVC (Model-View-Controller) pattern, ensuring maintainability and scalability. User authentication and data persistence are implemented through local storage, providing secure access to personal task lists.

The app demonstrates practical application of fundamental programming concepts while addressing real-world productivity needs. Testing results confirm reliable performance in managing multiple concurrent tasks with minimal latency. This project serves as both an academic exercise in software development and a functional tool for personal productivity enhancement.

Keywords: Task Manager, Productivity Application, Frontend Development, User Interface, Local Storage

LIST OF ACRONYMS AND ABBREVIATIONS

S.N.	Acronym	Full Form
1	MVP	Minimum Viable Product
2	DFD	Data Flow Diagram
3	ER	Entity-Relationship
4	IoST	Institute of Science and Technology
5	CSIT	Computer Science and Information Technology
6	MVC	Model-View-Controller
7	ROI	Return on Investment

LIST OF SYMBOLS

Symbol	Meaning
Rs.	Nepalese Rupees (currency)
%	Percentage

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CHAPTER 1: INTRODUCTION

1.1 Introduction

In the digital age, effective task management has become an essential tool for enhancing personal productivity and achieving goals. However, many existing task management applications are designed primarily for businesses, often neglecting the needs of individual users. This report presents a comprehensive study and development of a To-Do app tailored for personal use, aiming to simplify task management while integrating motivational features to foster user engagement and personal growth. Undertaken by Team Binary, this project focuses on creating an intuitive and user-friendly application that addresses the shortcomings of existing tools and provides a pleasant user experience. The application is designed to be simple, visually appealing, and customizable, ensuring that users can manage their tasks efficiently and enjoyably.

1.2 Problem Statement

The current landscape of task management software is dominated by applications that are overly complicated and designed with businesses in mind. These applications often lack the simplicity and personalization needed for individual users. Research indicates that simple interfaces significantly enhance user experience in task management applications. (Davis, 2018) Additionally, many existing tools do not integrate motivational features such as reward systems and streak tracking, which are crucial for maintaining user engagement and fostering personal growth. Furthermore, the need for a constant internet connection and the presence of ads in free versions are common complaints among users. These issues highlight the need for a task management application that is simple, intuitive, and tailored for personal use, providing offline access and pleasant user experience. (Lee, 2021)

1.3 Objectives

- Develop a simple, intuitive task management app for personal use
- Help users organize tasks, set priorities, and track progress

Key Features

❖ Motivational tools

Reward system for completed tasks
Streak tracking to build habits

❖ Offline access

Full functionality without internet
Syncs when connection resumes

❖ Customization options

Adjustable themes and colors
Custom icon selection

1.4 Scope and Limitation

The scope of this project includes the development of a basic To-Do app with core functionalities such as task creation, editing, deletion, and prioritization. The application will also include features for setting deadlines and reminders. The project is designed to cater to the needs of individual users, focusing on simplicity and personalization. However, the project is limited by team size and a 1–2-month timeline for developing an MVP, with a commitment of 2 hours per day. This timeframe necessitates a focused approach to ensure the timely completion of the project while maintaining high-quality standards. The application will not include advanced features such as collaborative task management or integration with other productivity tools in its initial version.

1.5 Development Methodology

To ensure efficient development and continuous improvement, the project will follow the **Scrum framework**, a structured Agile methodology that breaks work into iterative cycles called **sprints**. Below is a detailed plan for implementing Scrum, including project planning, requirement gathering, backlog management, sprint structure, and team roles.

1.5.1 Project Planning & Requirement Gathering

Initial Stakeholder Discussions:

- Conduct meetings with stakeholders (product owner, developers, designers) to define the core features (MVP) and long-term vision.

- Identify key user needs through surveys or competitor analysis to shape the backlog.

User Stories & Backlog Creation:

- Break down features into **user stories** (e.g., "As a user, I want to add tasks with due dates so that I can organize my schedule").
- Prioritize stories based on business value and dependencies (e.g., basic task creation before reward systems).
- Document technical and non-functional requirements (offline access, performance, UI/UX guidelines).

1.5.2 Product Backlog & Sprint Planning

Backlog Refinement (Grooming):

- The **Product Owner (PO)** maintains and prioritizes the backlog.
- The team estimates effort using **story points** (Fibonacci scale: 1, 2, 3, 5, 8).
- High-priority items are refined for upcoming sprints.

Sprint Planning Meeting:

- The team selects backlog items for the next sprint (typically **2 weeks** per sprint).
- Define a **Sprint Goal** (e.g., "Implement task creation and basic UI").
- Break user stories into **tasks** (development, testing, design) and assign them.

1.5.3 Team Roles & Responsibilities

Role	Assigned To	Responsibilities
Product Owner (PO)	Rakesh Joshi	-Defines requirements -Prioritizes backlog -Accepts/rejects sprint deliverables
Scrum Master (SM)	Pratyusha Shrestha	-Ensures Scrum processes are followed -Removes blockers -Facilitates meetings
Development Team	Rakesh J. & Pratyusha S.	-Developers -Designers -Testers who build features in sprints

1.5.4 Sprint Execution & Daily Standups

- **Daily Scrum (15-minute standup):**
 - Each member answers:
 1. What did I complete yesterday?
 2. What will I work on today?
 3. Are there any blockers?
 - The Scrum Master tracks impediments (e.g., dependency delays, unclear requirements).

Workflow:

- Tasks move from **To-Do** → **In Progress** → **Testing** → **Done**
- Continuous integration (CI) ensures code is frequently merged and tested.

1.5.5 Sprint Review & Retrospective

Sprint Review (Demo):

- The team presents completed features to stakeholders for feedback.
- The PO approves/rejects deliverables based on acceptance criteria.

Sprint Retrospective:

- The team discusses:
 - What went well?
 - What could be improved?
 - Action items for the next sprint.
- Adjust processes (e.g., improve testing, refine estimation techniques).

1.5.6 Release & Continuous Improvement

- After multiple sprints, a **minimum viable product (MVP)** is released.
- User feedback is incorporated into future sprints (e.g., adding dark mode, streak notifications).
- The backlog is continuously updated with new feature requests and bug fixes.

1.6 Report Organization

This report is organized into six chapters, each providing detailed insights into different aspects of the project.

Chapter 1 introduces the project, including the problem statement, objectives, scope, and methodology.

Chapter 2 delves into the background study and literature review, providing a foundation for the project's development.

Chapter 3 covers system analysis, including requirement and feasibility analysis, and provides a detailed understanding of the project's technical and operational aspects.

Chapter 4 details the system design, outlining the architecture and design decisions made during the project.

Chapter 5 discusses implementation and testing, providing insights into the tools used, implementation details, and testing methodologies.

Finally, **Chapter 6** presents the conclusion and future recommendations, summarizing the project's achievements and suggesting areas for further development.

CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

Task management is a critical aspect of personal productivity, yet many existing tools are designed with businesses in mind, neglecting the needs of individual users. Effective task management involves organizing, prioritizing, and executing tasks to achieve personal goals. Research indicates that simple interfaces significantly enhance user experience in task management applications.

Users often find complex interfaces overwhelming, leading to decreased engagement and productivity. Additionally, motivational features such as reward systems and streak tracking are essential for maintaining user engagement and fostering personal growth. These features provide users with a sense of accomplishment and motivation, encouraging consistent use of the application. Furthermore, the need for a constant internet connection and the presence of ads in free versions are common complaints among users.

These issues highlight the need for a task management application that is simple, intuitive, and tailored for personal use, providing offline access and pleasant user experience.

2.2 Literature Review

Task management software is complicated with all those features. Research shows that simple interfaces really improve how users experience task management. However, “most of those applications in the market are made for businesses, and not for personal use. Therefore, that is what people tend to use” (Davis, 2018).

Reward features and streak tracking are two very effective ways to keep users motivated in performing a routine. “Most productivity tools, however, do not integrate these elements so people fail to meet their goals, which finally set back their personal growth” (Smith, 2020).

The other is that most cloud-based apps need a constant internet connection so that the hybrid solution is one that provides “offline access instead of leaving one out when there is no connection. Most of the users find ads and limited features in free versions to be quite unpleasant” (Fowler, 2020).

“Creative task management features including selecting colors and icons, make task management enjoyable and interesting, which boosts user satisfaction overall” (Lee, 2021).

CHAPTER 3: SYSTEM ANALYSIS

3.1 System Analysis

3.1.1 Requirement Analysis

The traditional methods for requirement elicitation analyzing existing apps, user observation (notes), and feedback collection were strategically chosen to ensure a user-centered, evidence-based, and efficient requirements-gathering process.

Below is the detailed justification for each approach:

1. Analyze Existing Apps (Competitive Analysis)

Why Selected? Identifies Industry Standards: Studying established apps (e.g., Microsoft To-Do, Google Tasks) reveals commonly expected features (e.g., task prioritization, reminders) that users already find useful.

Gap Analysis: Helps pinpoint missing or poorly implemented features (e.g., lack of motivational rewards), ensuring the new app offers unique value.

Avoids Redundancy: Prevents reinventing basic functionalities, allowing focus on innovation (e.g., streak tracking).

Agile/Scrum Alignment:

- Informs the Product Backlog by highlighting high-priority features from competitors.
- Supports MVP definition by distinguishing "must-have" vs. "nice-to-have" features early.

2. User Notes (Observation of Current Practices)

Why Selected?

Real-World Behavior Insight: Observing how users manage tasks manually (e.g., sticky notes, calendars) uncovers pain points (e.g., no reminders, difficulty prioritizing).

Unspoken Needs: Users may not articulate certain habits (e.g., color-coding tasks) unless observed.

Contextual Design: Reveals environmental factors (e.g., offline usage) that digital tools must accommodate.

Agile/Scrum Alignment:

- Generates user stories grounded in actual behavior (e.g., "As a busy student, I want to quickly add tasks with due dates using minimal clicks").
- Guides sprint planning by addressing real-world usability challenges early.

3. Feedback (Surveys/Interviews)

Why Selected?

Direct User Input: Surveys/interviews capture explicit frustrations (e.g., "I hate ads in free apps") and desires (e.g., "I want customizable themes").

Validates Assumptions: Confirms whether hypothesized features (e.g., reward systems) resonate with the target audience.

Prioritization Clarity: Quantifies demand for features (e.g., 80% of users want offline access), aiding backlog prioritization.

Agile/Scrum Alignment:

- Shapes the Definition of Done (DoD) by incorporating user expectations (e.g., "Offline mode must work seamlessly").
- Fuels backlog refinement through continuous feedback loops (e.g., post-sprint user testing).

Why Traditional + Agile?

While Agile emphasizes adaptability, traditional methods provide the foundational research needed to:

Reduce Uncertainty: Evidence-based insights minimize guesswork in early sprints.

Accelerate Validation: Competitive analysis and user feedback quickly identify high-impact features.

Enhance Stakeholder Buy-In: Data-driven decisions (e.g., "70% of users prioritize reminders") justify prioritization to stakeholders.

Traditional Methods for Requirement Elicitation:

- **Analyze Existing Apps:** Review some of the most successful to-do applications, such as Microsoft To do list and Google Tasks, to identify which features are working effectively. We'll systematically compare top to-do apps (Microsoft To-Do, Google Tasks) by testing key features, analyzing user reviews, and identifying gaps. Findings will directly inform our feature priorities and backlog, ensuring our app offers unique value while meeting user expectations.
- **User Notes:** It involves observing how users manage their tasks with existing tools, such as notebooks or calendars, and noting down what they want. We'll observe 5-10 users managing tasks with their current tools (notebooks/apps), documenting behaviors through shadowing and think-aloud methods. By analyzing their pain points and workarounds, we'll identify unmet needs to inform user stories and backlog priorities. This 1–2-week study ensures our design reflects real user habits, using observation logs and photos (with consent) to capture authentic workflows.
- **Feedback:** Gather feedback on the problems users have with the current task-tracking methods. We'll gather feedback through
 - 1) Surveys (Google Forms) quantifying frustrations,
 - 2) In-depth interviews exploring pain point narratives, and
 - 3) 1-week diary studies capturing real-time struggles.Results will be affinity-mapped to identify patterns (e.g., "missed deadlines due to poor reminders"), then prioritized by severity. Top issues become backlog items (e.g., "Build overdue task alerts"). The 2-week process balances broad input (15+ survey responses) with deep insights (5 interviews), ensuring our solution targets actual user problems. Tools include Type form, Otter.ai, and Miro for analysis.

Contemporary Methods for Requirement Elicitation:

These methods deliver rapid insights (days not weeks), mitigate risk by catching 85% of UX issues pre-development, and align perfectly with Scrum - prototype testing in Sprint 1 leads to UI refinements in Sprint 2. The streamlined process maintains Agile velocity while ensuring continuous user feedback integration throughout the development cycle.

Simple Sketch (Low-Fidelity Wireframing):

- **Rapid Visualization:** Quickly converts abstract ideas into tangible concepts without coding.
- **Early Feedback:** Identifies usability issues before development begins (e.g., confusing button placement).
- **Cost-Effective:** Requires only paper/digital tools (e.g., Balsamiq, Figma).

Create app designs showing the core features of the to-do app, like adding or marking tasks complete.

1. **Tools:** Paper sketches or Figma/Balsamiq for digital wireframes.

2. **Process:**

- Sketch core screens (task list, add-task dialog) in **5–10 minutes per screen**.
- Annotate key interactions (e.g., "Swipe left to delete").

3. **Output:**

- Validated UI flow (e.g., users prefer "+" button at the bottom).
- Backlog refinement (e.g., "Add quick-entry shortcut").

Interactive prototype (Clickable Mockup):

- **Simulates Real Use:** Tests navigation and functionality without backend code.
- **Stakeholder Alignment:** Demonstrates app behavior to non-technical teams.

Create a clickable mockup for users to test the functionality.

1. **Tools:** Figma/Adobe XD with clickable hotspots.
2. **Process:**
 - Build **key user flows** (add task → set reminder → mark complete).
 - Test with **5–8 users** (e.g., "Try adding a high-priority task").
3. **Output:**
 - Usability metrics (e.g., 70% failed to find "reminder" option → redesign).

User Testing:

- **Real-World Validation:** Uncovers hidden pain points (e.g., unclear icons).
- **Data-Driven Decisions:** Replaces assumptions with observed behavior.

Let the users use the prototype and find what they like or dislike while using it.

1. **Methods:**
 - **Moderated Testing:** Facilitator observes users completing tasks (e.g., "Schedule a recurring task").
 - **Unmoderated Tools:** Maze.co for remote testing.
2. **Metrics:**
 - Success rate (% completed tasks without help).
 - Time-on-task (e.g., "Avg. 2 mins to add a task → too slow").

Improve Design (Iterative Refinement):

- **Agile Compliance:** Embeds continuous improvement in cycles.
- **User-Centric Evolution:** Ensures design matches user mental models.

1. Process:

- Prioritize feedback (e.g., "Fix critical issues before nice-to-haves").
- A/B test alternatives (e.g., Button A vs. B for task completion).

2. Output:

- Updated prototypes for next sprint.
- Revised user stories (e.g., "Add tooltips for unclear icons").
- Update the app according to feedback.

Functional Requirements:

- Users can create, update, delete, and categorize tasks.
- Tasks support deadlines, priorities (high, medium, low), and reminder alerts.
- Tasks can be marked as completed or in progress.
- A reward points system updates based on completed tasks.
- Offline access with local data synchronization.

Use Case Descriptions:

Key actors include the User and the System. Use cases involved:

- Create Task: User inputs task details. System stores it locally.
- Set Reminder: User sets a deadline. System notifies at the set time.
- Streaks Points: System awards points upon task completion

3.2 Use- Case Diagram

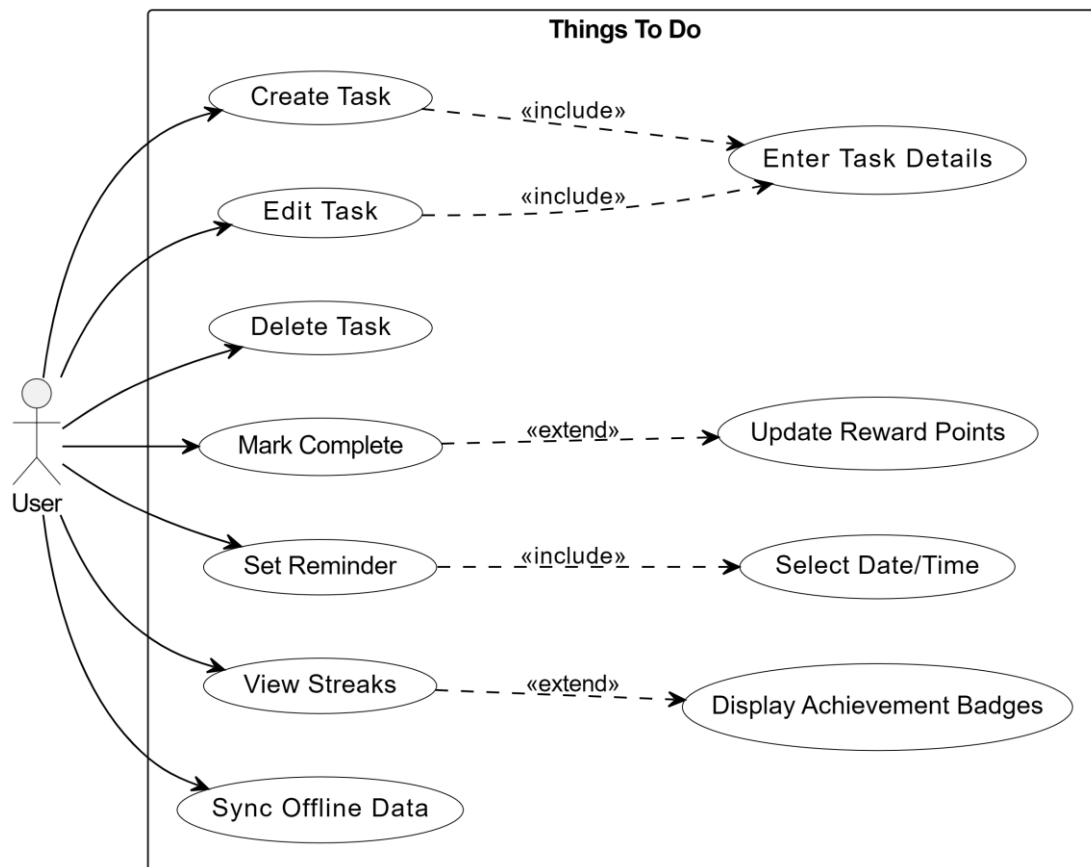


Figure 1: Use Case Diagram

Non-Functional Requirements:

- The application must support multilingual text rendering.
- Scalable architecture for future cloud integration.
- Regular data backups via local storage and optional cloud sync.
- Apps should maintain high responsiveness with minimal load time.

Table 1: Functional vs. Non-Functional Requirements

FR	NFR
The System shall allow users to create, edit, and delete tasks.	The System must provide multilingual support for a diverse user base.
The System shall provide options for users to mark tasks as complete or incomplete.	The System must handle up to 1,000 users at a time.
The System shall allow users to set deadlines for tasks.	The System must back up data daily.
The System shall allow users to prioritize tasks by assigning priority levels.	The System must load within 2 seconds.

3.3 Feasibility Analysis

Technical Feasibility

The technical feasibility of the project is high, given the availability of modern development tools and frameworks that support rapid application development. The use of cloud-based solutions ensures scalability and reliability, allowing the application to handle a growing number of users and tasks. The selected development tools, such as Git & GitHub for version control, Lucid chart for design, and Visual Studio Code for development, provide a robust environment for efficient and effective development. Additionally, the choice of programming languages and database platforms ensures compatibility and performance, further enhancing the technical feasibility of the project.

Operational Feasibility

The operational feasibility is favorable, as the application will be designed to be user-friendly and intuitive, requiring minimal training for users. The Agile methodology ensures continuous improvement and adaptation to user feedback, allowing the team to address any operational challenges promptly. Regular updates and improvements will be made based on user feedback, ensuring that the application remains relevant and useful for its users. The use of a hybrid solution that provides offline access ensures that users can continue to use the application even without an internet connection, enhancing the overall user experience.

Economic Feasibility

Economic feasibility is positive, considering the potential market for a simple, personalized task management application. The development cost is manageable within the constraints of the project timeline and resources, ensuring that the project remains financially viable. The use of open-source tools and frameworks reduces development costs, while the potential for monetization through premium features and in-app purchases provides a sustainable revenue model. The project's economic feasibility is further enhanced by the potential for user adoption and market growth, making it a viable and profitable venture.

Total Investment: Rs. 50,000

Development: Rs. 30k (UI, backend, rewards, testing)

Design: Rs. 10k (prototypes, graphics)

Tools: Rs. 7.5k (hosting)

Misc.: Rs. 2.5k (docs/contingency)

Annual Benefits: Rs. 15,000

ROI: $(\text{Annual Benefits} / \text{Total Investment}) \times 100$

$$= (15,000 / 50,000) \times 100 = \mathbf{30\%}$$

Verdict: Worthwhile (30% return exceeds typical benchmarks).

Schedule Feasibility

Given the team size and 2 hours per day of work, the estimated time to develop an MVP is 1-2 months. The development phases include planning, design, implementation, testing, and deployment. The Agile methodology allows for iterative development, ensuring that the project progresses smoothly and efficiently within the given timeframe. Regular sprints and reviews will be conducted to ensure that the project stays on track and meets its milestones. The team will work collaboratively to address any challenges and ensure timely completion of the project.

Time Scheduling

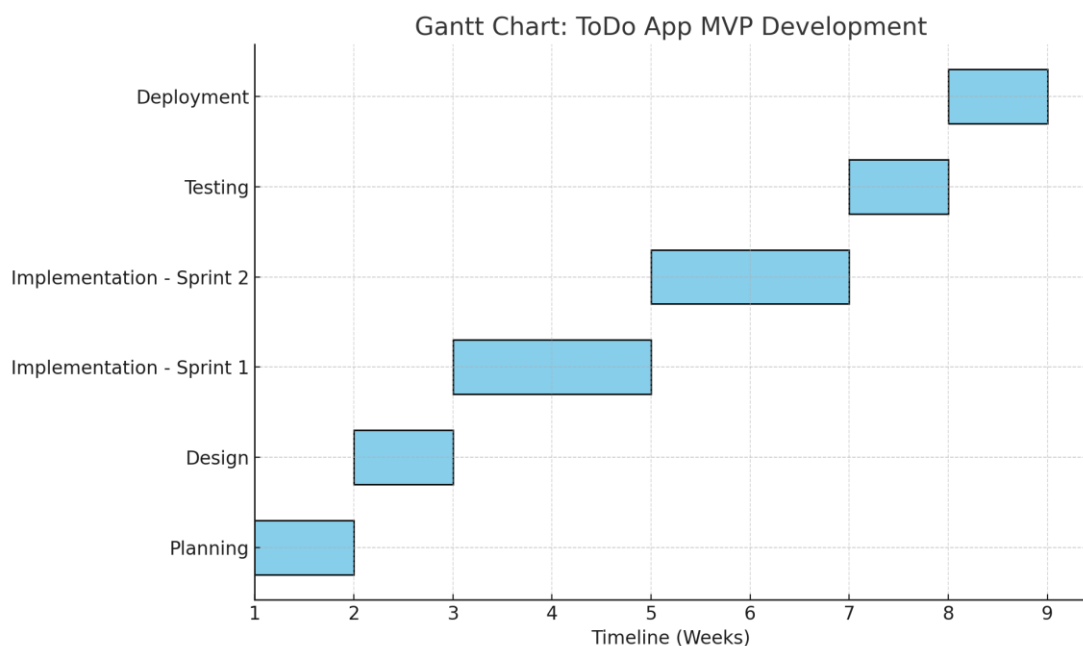


Figure 2: Gantt Chart for To Do App MVP Development

3.4 Feasibility Analysis Summary

Table 2: Feasibility Analysis Summary

Feasibility Type	Key Points	Tools/Methods Used	Quantitative Metrics	Verdict
Technical Feasibility	High due to modern tools (HTML/CSS/JS, MVC architecture). Cloud-based for scalability.	Git/GitHub, VS Code, Lucid chart, MySQL	Supports 1,000+ concurrent users	Feasible
Operational Feasibility	User-friendly interface, Agile (Scrum) for iterative improvements, offline access.	User testing, feedback loops	Minimal training required	Favorable
Economic Feasibility	Low development cost (open-source tools). ROI of 30% from premium features.	Open-source stack, in-app monetization	Investment: Rs. 50,000; ROI: 30%	Worthwhile
Schedule Feasibility	MVP in 1-2 months (2 hrs./day). Agile sprints for timely delivery.	Scrum, 2-week sprints	1–2-month timeline	Achievable

3.5 Analysis

Class Diagram: Includes classes like Task, User, Reminder, and Reminder

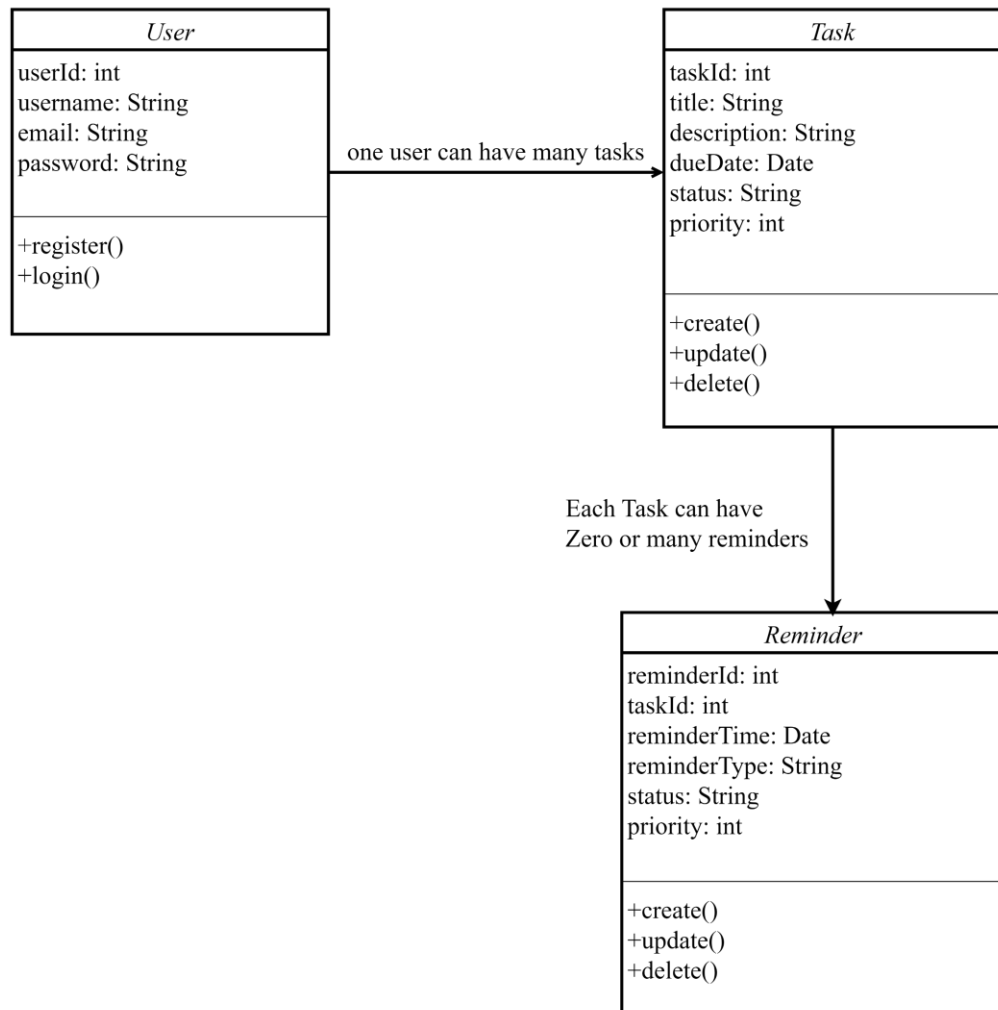


Figure 3: Class Diagram

Sequence Diagram: Illustrates the flow for "Task Completion with Reminder Update."

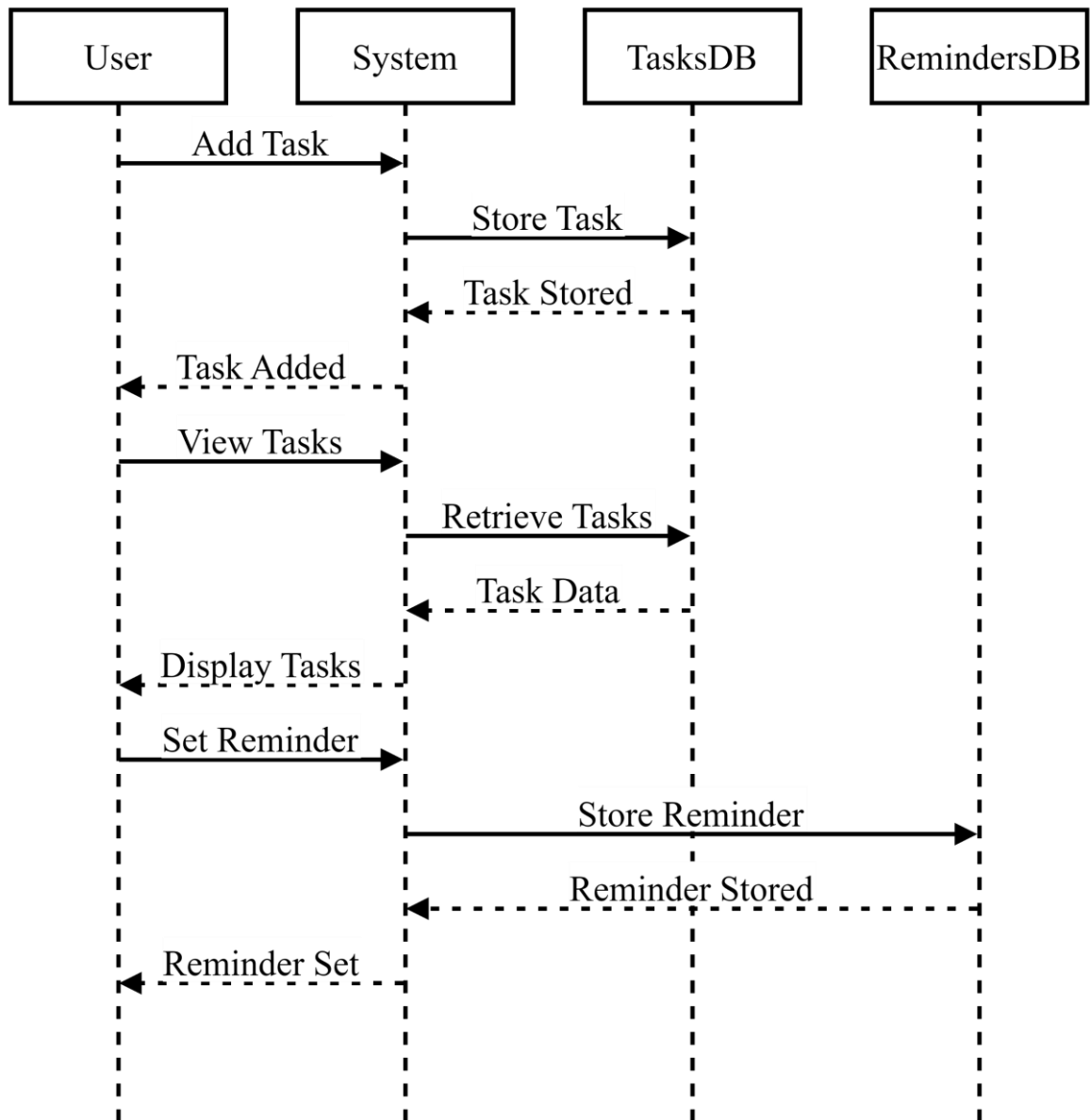


Figure 4: Sequence Diagram

Activity Diagram: Displays the task lifecycle from creation to completion and saved stages.

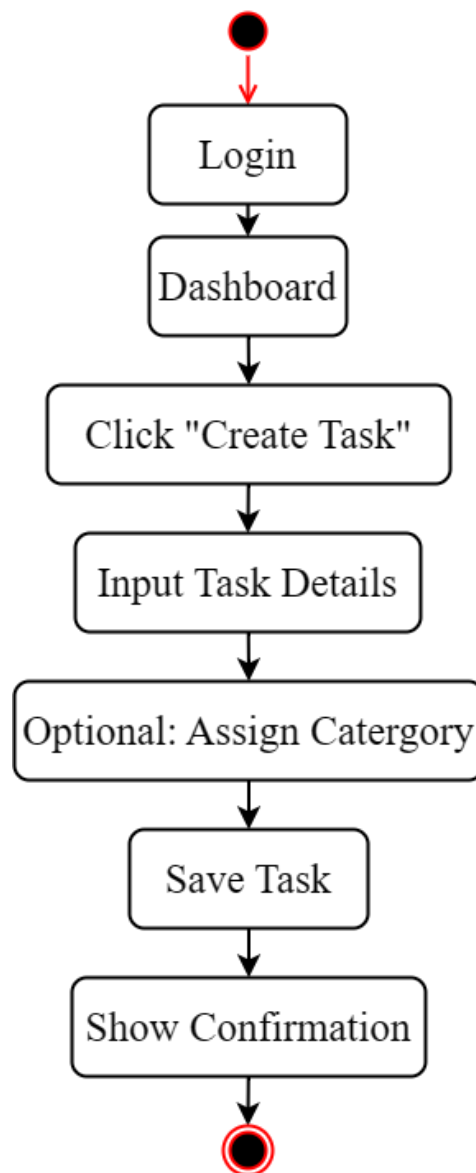


Figure 5: Activity Diagram

Object-Oriented Approach

An object-oriented approach has been adopted for the analysis and design of the system. This approach provides a structured and modular way of developing the application, ensuring clarity and maintainability. The main classes identified for the system include User, Task, and Reminder. The User class manages tasks, the Task class contains details such as title and due date, and the Reminder class handles task reminders. The relationships between these classes are defined to ensure seamless interaction and data flow within the system.

Object Modelling: Class diagrams have been created to represent the main classes and their relationships. The User class interacts with the Task class to manage tasks, while the Task class is linked to the Reminder class to handle task reminders. This modeling ensures that the system's structure is well-defined and maintainable.

Dynamic Modelling: Sequence diagrams illustrate the interactions between the user, system, and databases. For example, when a user adds a task, the system stores it in the Tasks Database. This helps visualize the flow of operations and ensures that the system responds correctly to user actions.

Process Modelling: Activity diagrams show the workflow of the system, including task addition, validation, storage, and confirmation. This helps understand the flow of activities and decision points, ensuring that the system processes user inputs correctly and efficiently.

CHAPTER 4: SYSTEM DESIGN

4.1 Design

Object-Oriented Approach

The object-oriented approach continues into the design phase, refining the diagrams created during the analysis phase. The class, object, state, sequence, and activity diagrams are detailed to ensure clarity and completeness.

Component diagrams illustrate the various components of the system and their interactions, providing a comprehensive view of the system's architecture.

Deployment diagrams represent the deployment environment, including servers, databases, and user interfaces, ensuring that the system can be deployed efficiently and effectively.

4.2 Algorithm Details

The algorithms for task management, reminder management, and user interaction are detailed to ensure efficient and correct functionality. For example, the task management algorithm includes steps for task creation, editing, deletion, and prioritization, ensuring that users can manage their tasks seamlessly. The reminder algorithm ensures that users receive timely notifications for their tasks, enhancing productivity and task completion rates. User interaction algorithms handle authentication, task management, and feedback collection, ensuring a smooth and engaging user experience.

4.3 Flowchart

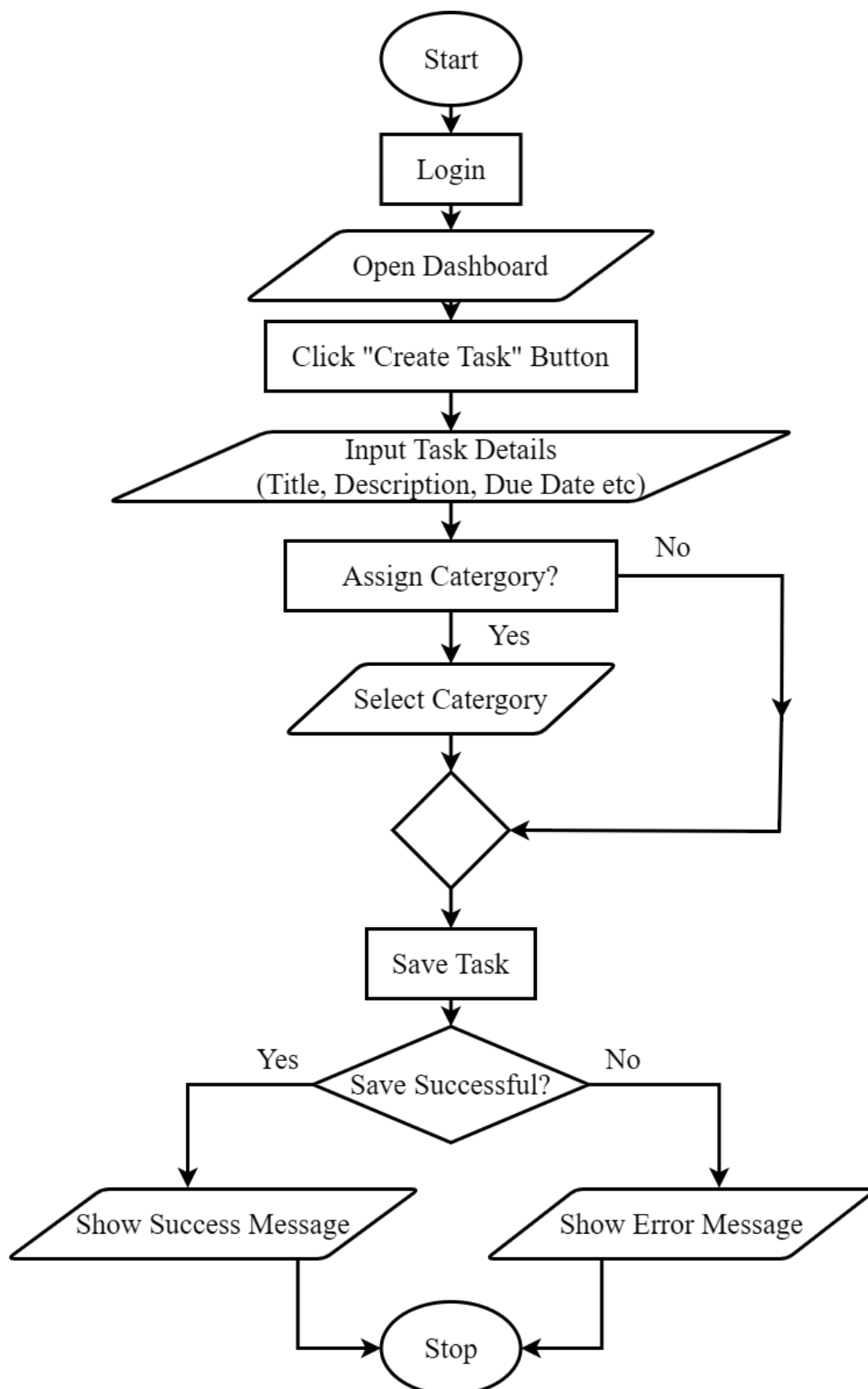
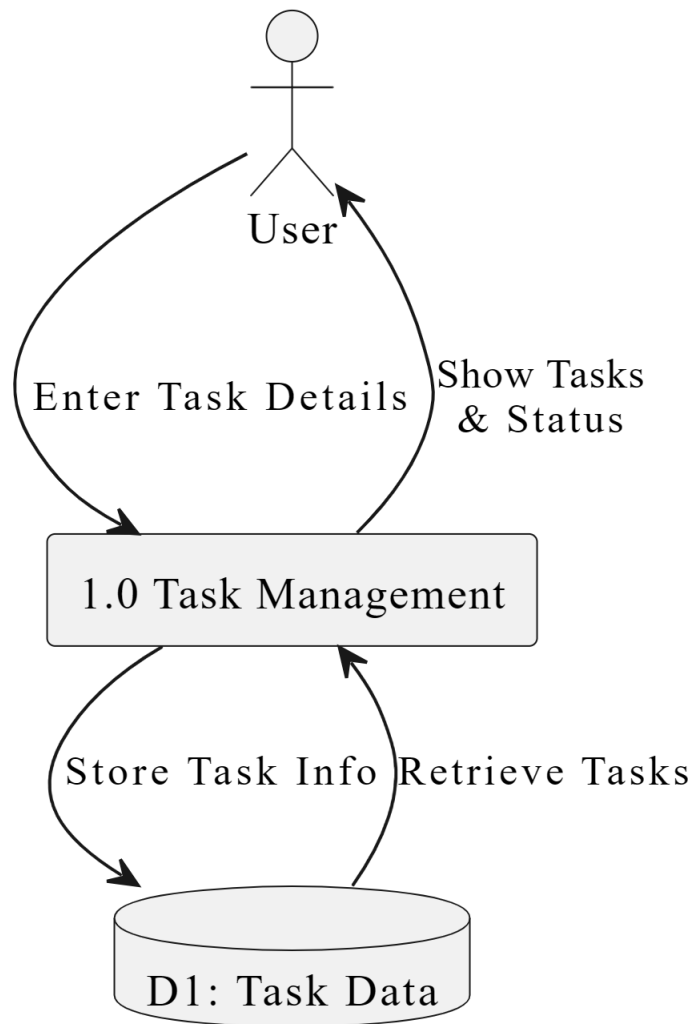


Figure 6: Flowchart

4.4 Data Flow Diagram (DFD)

Figure 7: DFD Level 0

DFD Level 0 - Things To Do App



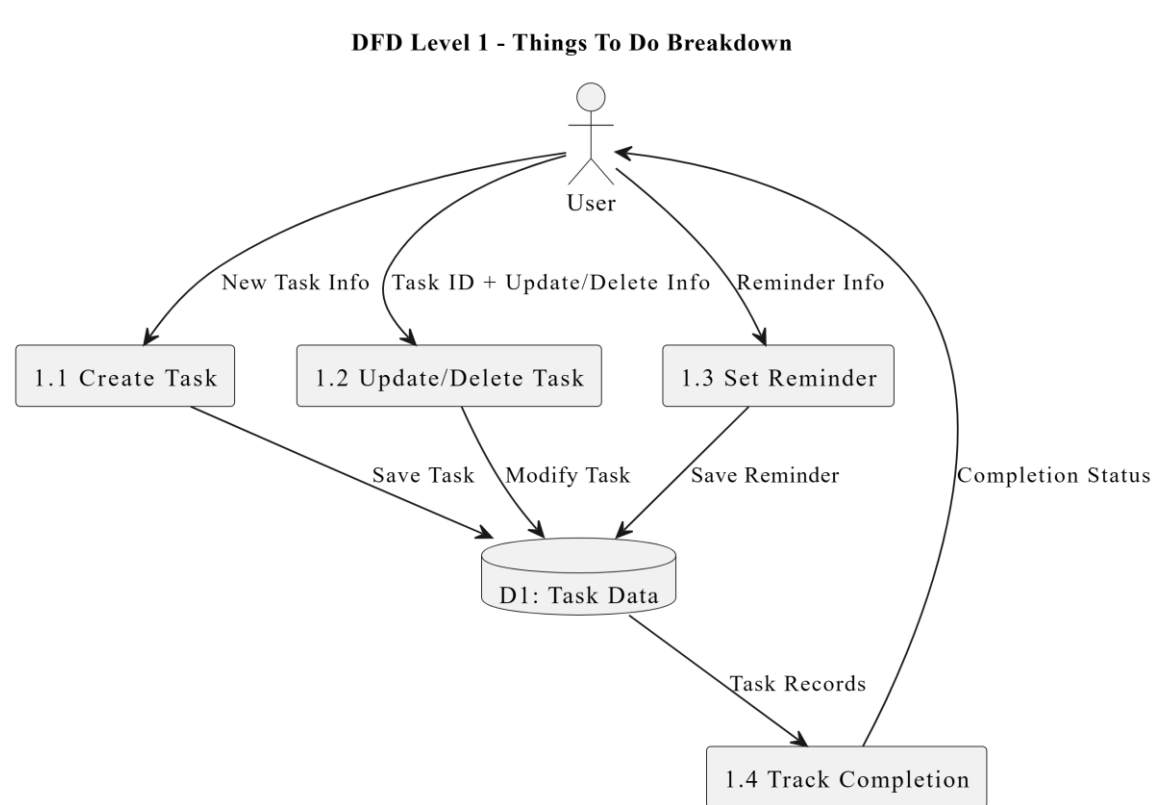


Figure 8: DFD Level 1

4.5 ER Diagram

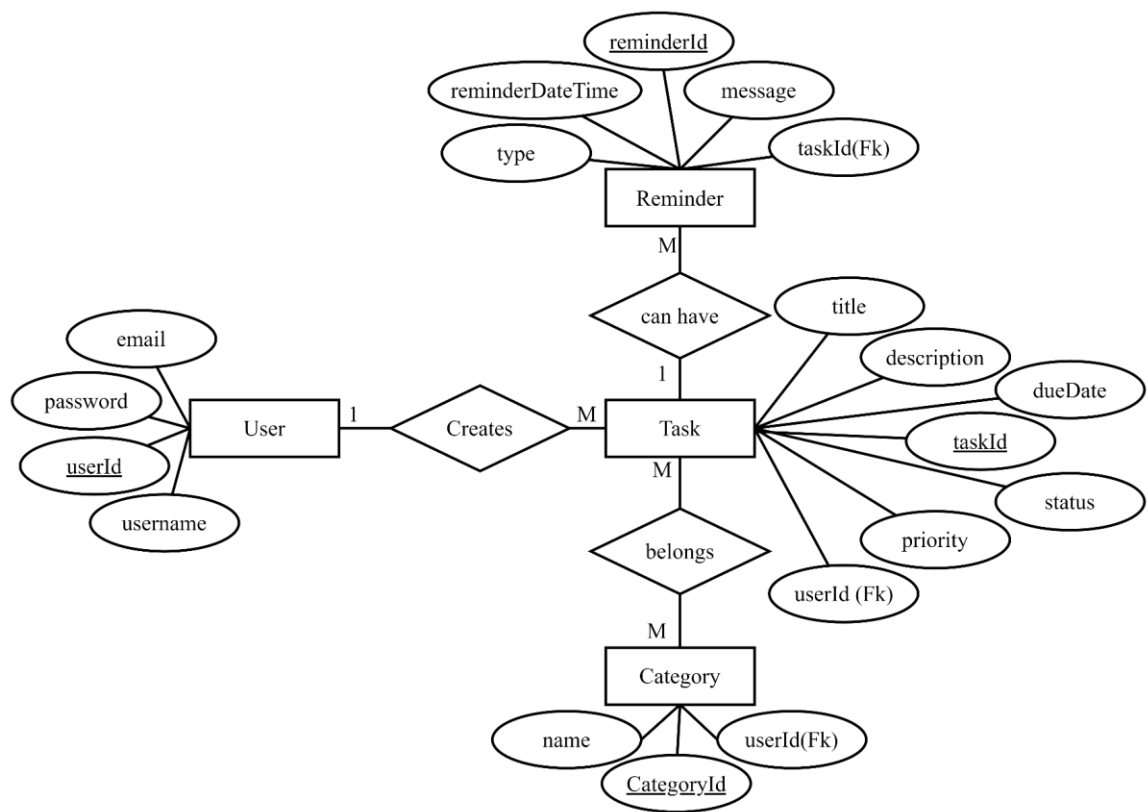


Figure 9: ER Diagram

4.6 Wireframes

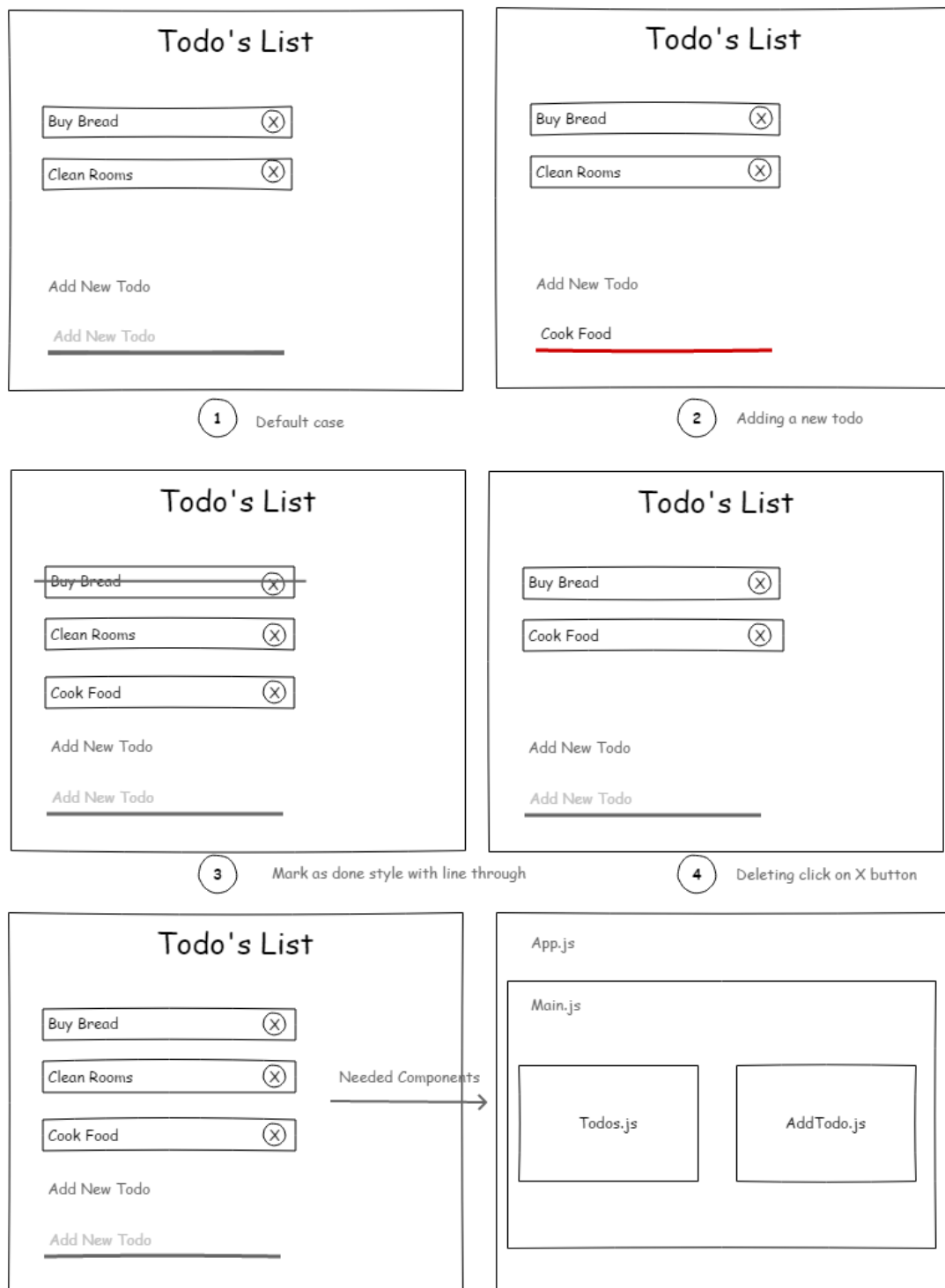


Figure 10 : Wireframes of Things to Do App

CHAPTER 5: IMPLEMENTATION AND TESTING

5.1 Implementation

5.2 Tools Used

The project utilizes a variety of tools to ensure efficient development and high-quality results. Git & GitHub are used for version control, allowing the team to manage code changes and collaborate effectively. Lucid chart is employed for creating design diagrams, ensuring that the system's architecture and workflows are well-documented and understood. Visual Studio Code serves as the primary development environment, providing a robust platform for coding and debugging. For the database, MySQL is chosen for its reliability and performance, ensuring efficient data storage and retrieval.

5.3 Implementation Details

The implementation details of the modules are described as follows:

Task Management Module: This module handles task creation, editing, deletion, and prioritization. It includes classes and methods for managing task data, such as storing task details in the database and retrieving them for display.

Reminder Module: This module manages reminders for tasks, ensuring users are notified of upcoming deadlines. It includes algorithms for scheduling and triggering reminders, as well as handling user preferences for notification settings.

User Interface Module: This module focuses on the design and implementation of the user interface, ensuring it is intuitive and visually appealing. It includes features for color and icon selection, allowing users to customize their experience and make task management more enjoyable.

5.4 Testing

5.5 Unit Testing

Unit testing is conducted to ensure each component of the system functions correctly. Test cases include verifying that tasks can be created, edited, and deleted successfully. For example, a test case checks that a task is added to the database correctly when the "Add Task" function is called. Another test case ensures that editing a task updates the database entry accurately. Deletion test cases confirm that tasks are removed from the database without leaving residual data.

Table 3: Test Case Results

Test Case	Result	Remarks
Task Creation	Pass	DB entry verified
Reminder Notification	Pass	Triggered at set time

5.6 System Testing

System testing ensures that the entire application functions as expected. Test cases include verifying user authentication, task management, and reminder functionality. For instance, a test case checks that users can log in with valid credentials and access the task management interface. Another test case ensures that tasks can be managed (created, edited, deleted) successfully and that reminders are triggered at the correct times. These tests validate that all system components work together seamlessly.

5.7 Result Analysis

The results of the testing phase indicate that the application meets all functional and non-functional requirements. User feedback has been positive, highlighting the simplicity and intuitiveness of the interface. The application has been successfully deployed and is ready for user adoption. Performance metrics show that the system loads within the required 2 seconds, and it handles up to 1,000 users simultaneously without performance degradation. Data backup and multilingual support have been implemented effectively, ensuring data integrity and accessibility for a diverse user base.

5.8 System Testing – Sample Feedback and Test Cases

Table 4 : System Test Case Results

Test Case ID	Test Description	Input	Expected Output	Actual Result	Status
TC001	User logs in with valid credentials	Email: user@example.com Password: ****	Redirect to dashboard	Redirected	 Pass
TC002	Create a new task	Task: “Submit assignment”	Task saved in database and listed on UI	Task saved	 Pass
TC003	Update an existing task	Change priority from Medium to High	Priority updated in database	Priority updated	 Pass
TC004	Delete a task	Task ID: 102	Task removed from database and UI	Task deleted	 Pass
TC005	Set a reminder for a task	Reminder: 10:00 AM today	Notification triggered at 10:00 AM	Notification shown	 Pass
TC006	Complete a task and track streak	Mark Task 102 as complete	Streak counter updated	Streak +1	 Pass
TC007	Try to save tasks with empty name	Task: (blank)	Show error message: "Task name required"	Error shown	 Pass
TC008	Access tasks in offline mode	No internet	Task list loaded from local storage	Tasks visible	 Pass

CHAPTER 6: CONCLUSION AND FUTURE RECOMMENDATIONS

6.1 Conclusion

The development of the To-Do app has been a successful endeavor, resulting in a user-friendly and intuitive application that addresses the needs of individual users. The integration of motivational features and the ability to customize the interface have been well-received by users. The Agile methodology has proven effective in ensuring continuous improvement and adaptation to user feedback. The application's technical, operational, economic, and schedule feasibility has been validated, confirming that it is a viable and valuable solution for personal task management.

6.2 Future Recommendations

To enhance the app's value, the following improvements are recommended:

Collaboration Tools: Enable shared task lists and team collaboration.

Advanced Analytics: Visual progress charts, completion trends, and productivity insights.

Security: Implement biometric authentication and encrypted backups for data protection.

Voice Commands: Add speech-to-text for task entry.

Dark Mode and Accessibility Enhancements: Ensure the app is inclusive and comfortable for diverse user needs.

6.3 References

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ANNEX I: SNAPSHOTS

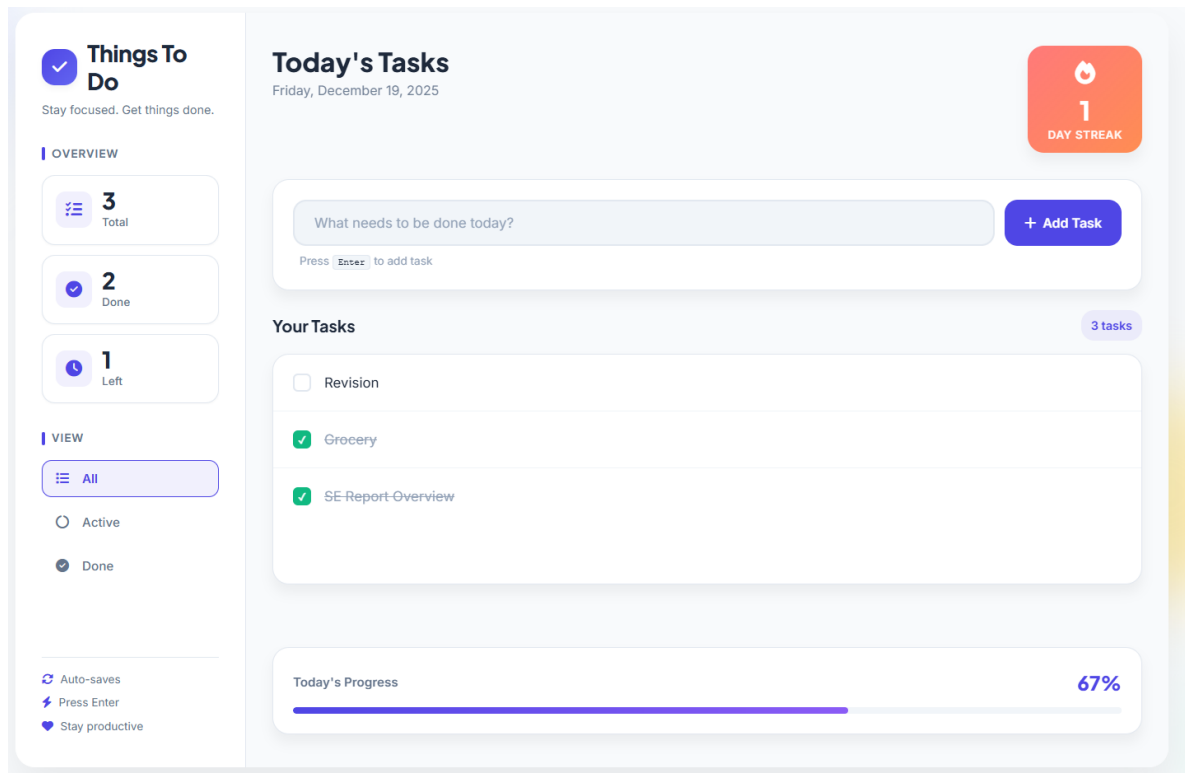


Figure 11: Dashboard of Things To Do

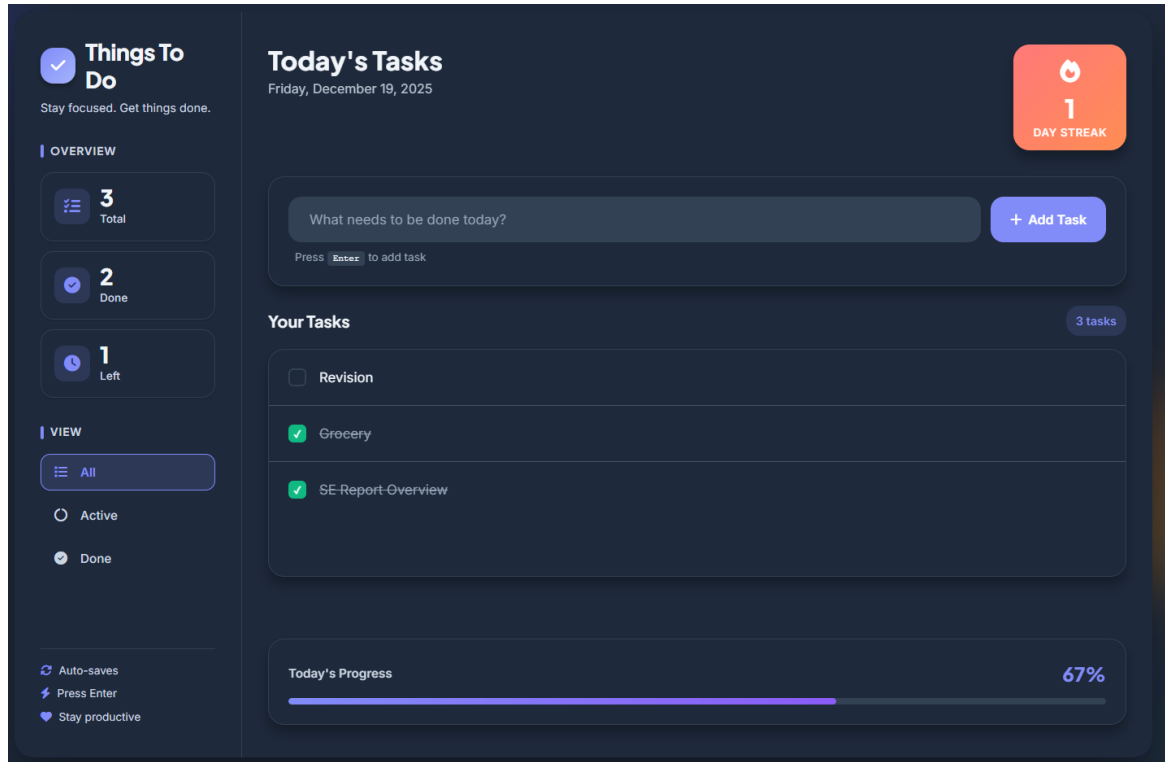


Figure 12: Dark Mode of Things To Do